12th hacker series

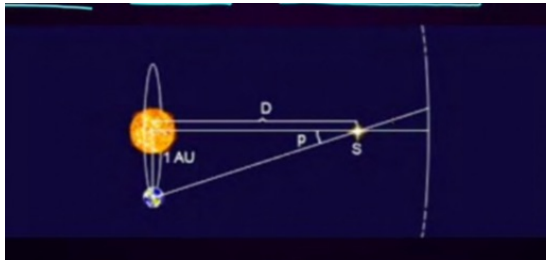
Evolution

Evolution

- evolution is the changes in the flora and fauna that took place gradually over millions of years due to cumulative changes in organisms

Stellar distance

- Distance between earth and a star is called stellar distance.
- These distances can be measured in 'light years'
- One light year is the distance travelled by light in one year in vacuum
- When we look at stars , in a way we are peeping into the past



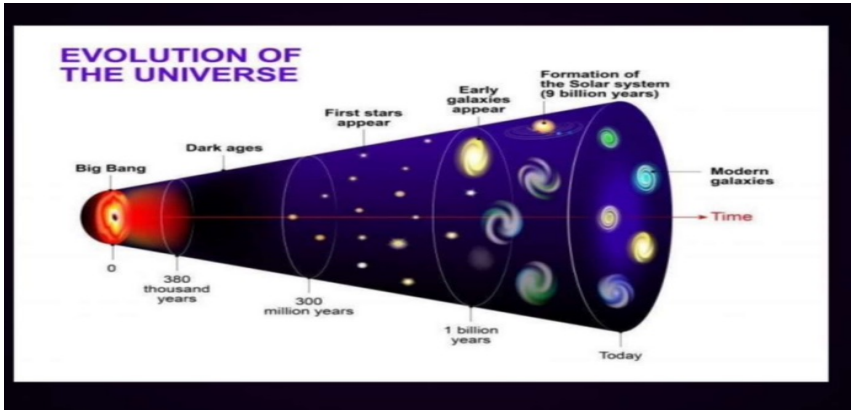
Origin of universe

- 20 billion years ago
- Big Bang theory explains this



Events after expansion of universe

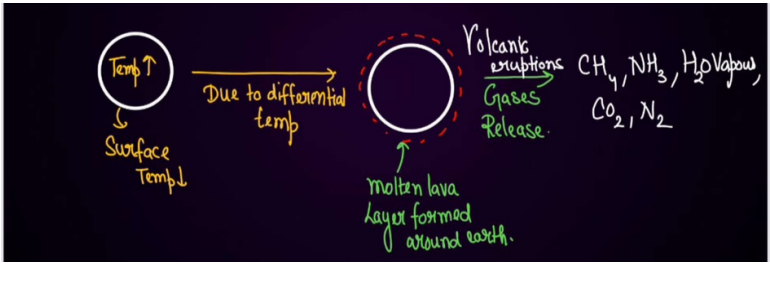
- temperature declined
- H2 + He formed
- Gases condensed
- Galaxies of present day formed





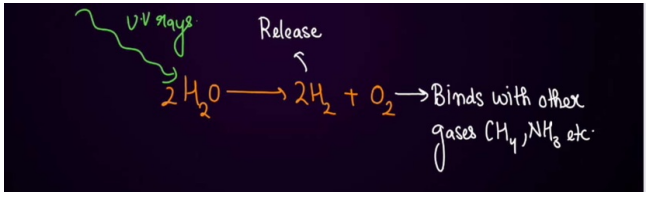
Origin of earth

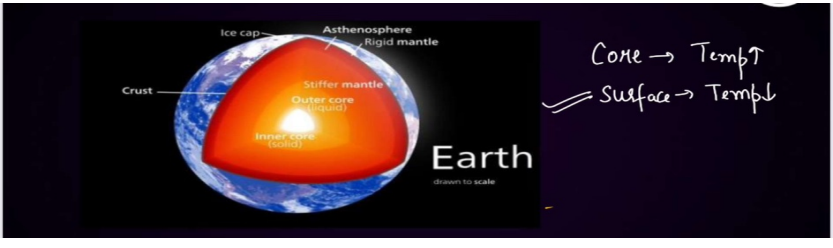
4.5 billion years ago

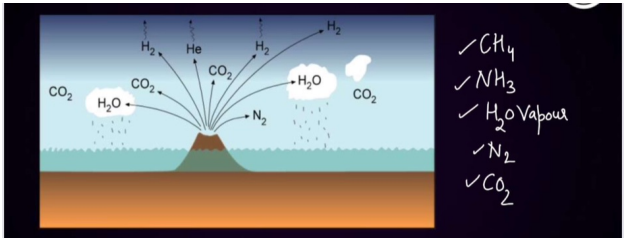


Primitive earth

- no atmosphere
- Reducing (initially)
- High temperature
- no ozone. ⟶ UV rays could enter easily.







Theory of origin of life

Theory is given by various scientist on how life originated

Theory of special creation

- given by father Suarez
- Life was created by some supernatural power: God
- life was created as such in six days
- Diversity was created as such and will remain like this
- Earth is 4000 year old

Theory of catastrophism

- catastrophic events (calamities) , wipes of the existing life and new life comes

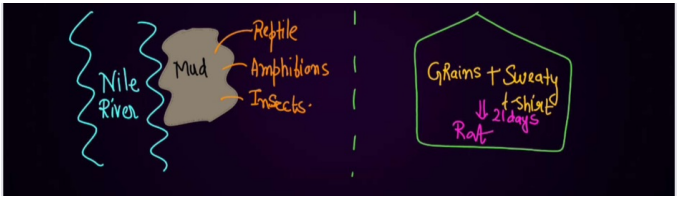
Extraterrestrial theory- theory of panspermia/ cosmozoic theory

- life came from outer space in the form of seed/spore etc



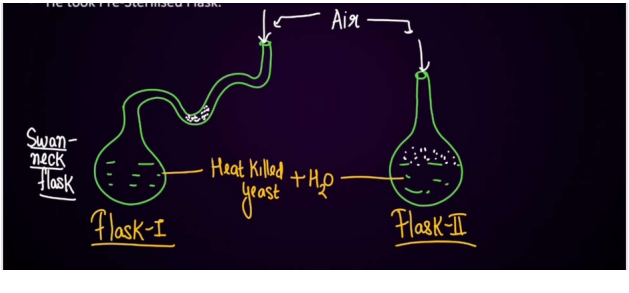
Theory of abiogenesis/ theory spontaneous generation

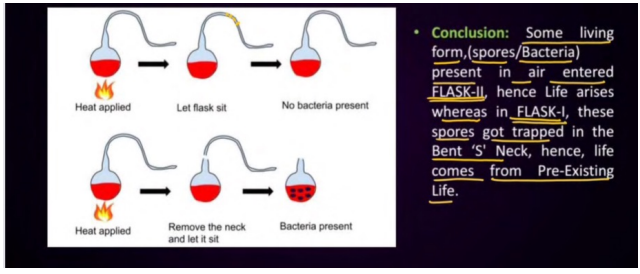
- life comes from nonliving things like mud, straw, Grain, etc



Theory of biogenesis

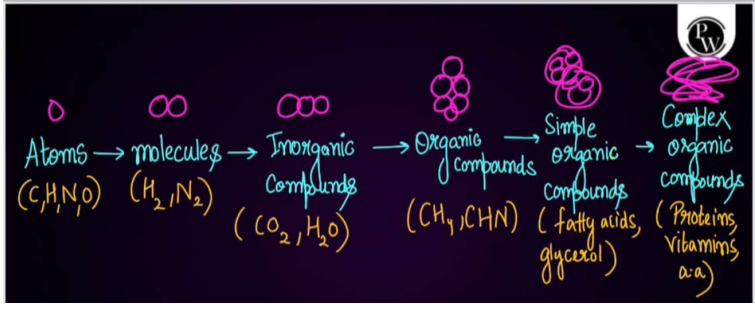
- life comes from pre-existing life
- Pasteur's experiment to prove biogenesis that rejected theory of abiogenesis
- he took pre-sterilised flask

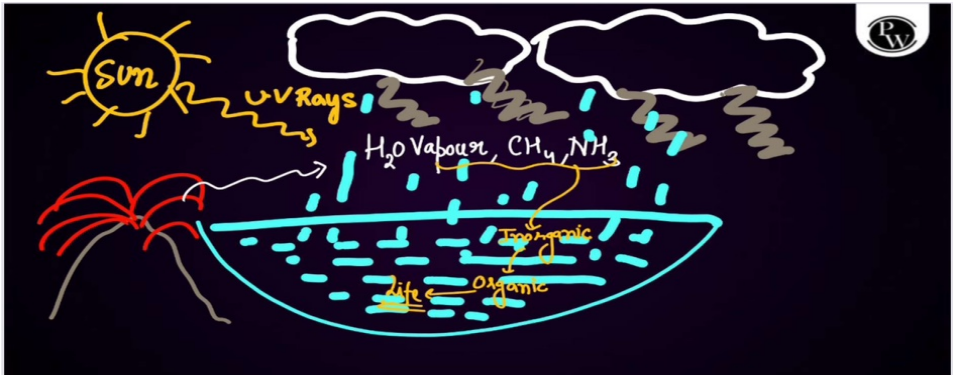




Theory of chemical evolution

- most widely accepted theory of origin of life
- this theory was proposed by oparin and haldane
- They said primitive atmosphere on earth, supported origin of life like:
- 1) reducing atmosphere
- 2) lightning
- 3) high temperature- sun uv rays, volcanic eruption





Eventually, later ozone was also formed, temperature, reduced and water vapour eventually condensed, and ultimately fell on earth, depression as rain (rainfall occurred for million years)

↓

Forms the ocean, which we see today

↓

Water was hot with various chemicals

↓

First life probably came

↓

Life probably came 500 million years after formation of earth i.e, approx 4 bya

QUESTION

The origin of life according to the early Greek philosophers was transfer of unit of life from outer space to the different planets in the form of (2024)

A seeds

B spores

C gemmules

D gametes.



QUESTION

The theory of evolution supported by the experiment conducted by Louis Pasteur is (2020)

A Spontaneous generation theory

B Life comes only from pre-existing life

C Abiogenesis of life

D Big bang theory



QUESTION

Louis Pasteur demonstrated that (2020)

A early life came from outer space

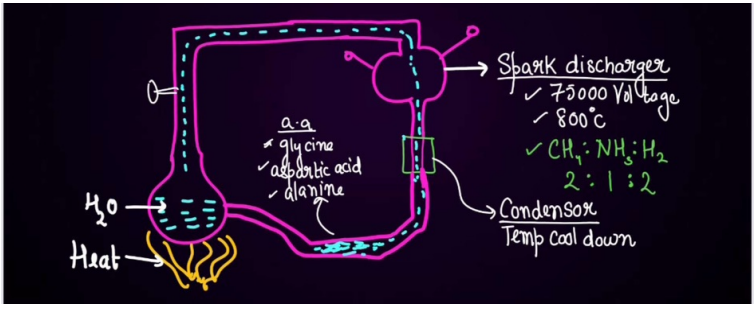
B non-living chemicals produced living molecules

C life comes from pre-existing life

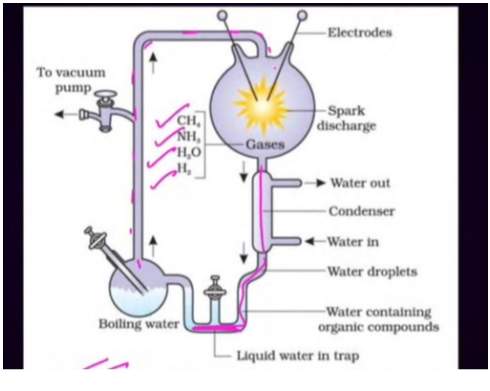
D life originated spontaneously.

Miller and Urey experiment

- in 1953
- Performed, experiment to prove theory of chemical evolution
- he took ammonia, CH4, H2 along with water vapour at 75k voltage 800°C temperature in spark discharge Chamber
- After 18 days of experiment, they collected the water and found simple amino acids like glycine, aspartic acid and alanine
- Tube of Spark discharge chamber(hot with gases). ⟶ Passed through. (condenser.). ⟶ Cooling.⟶ Water trap.



- chemical evolution: nonliving organic molecules- 1st life came
- First non-cellular form of life- 3 bya
- First cellular form of life- 2 bya
- First life in water



QUESTION

State two postulates of Oparin and Haldane with reference to origin of life.

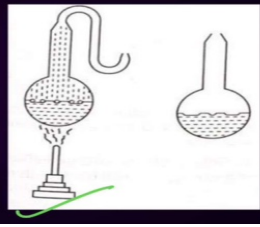
(AI 2017)

- ✓ No atmosphere
- ✓ Reducing envi
- ✓ High Temp.
- ✓ no ozone

QUESTION

Given below is an illustration of the experiment performed by Louis Pasteur. Explain the experiment and its conclusion. (2024)

- ✓ Explain theory of biogenesis
- ✓ Conclusion



QUESTION

(a) Explain the significance of the experiment carried out by S.L. Miller. Name the scientists whose hypothesis prompted him to carry out this experiment.

(b) How does meteorite analysis favour this hypothesis?

(2024)

a) Oparin & Haldane.
↓
Theory of Chemical evol
↓
proof → Miller & Urey

b) Meteorites analyse → Same content as organic molecules

QUESTION

State Oparin and Haldane hypothesis. How did S.L. Miller experimentally prove it? Explain. (2020 C)

Describe the experiment that helped Louis Pasteur to dismiss the theory of spontaneous generation of life. (Delhi 2016)

Theories for evolution: biological/ organic evolution

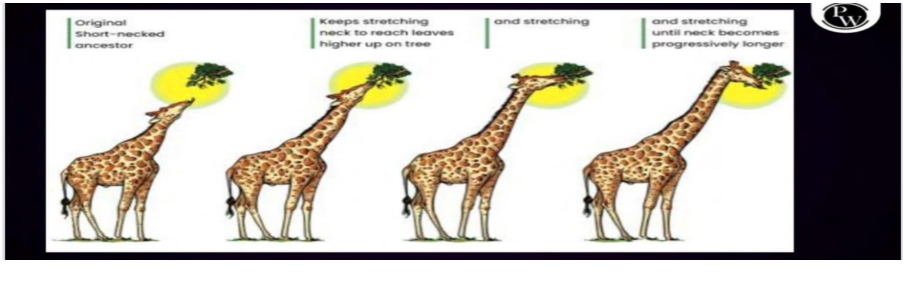
How do variations occur occurred in species?

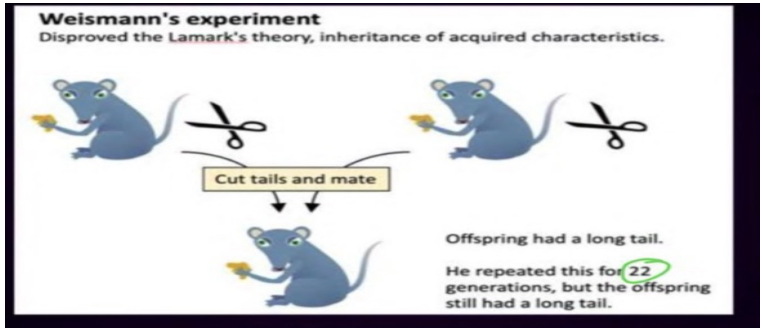
Lamarckism/ theory of use or disuse of organ/theory of inheritance of acquired character

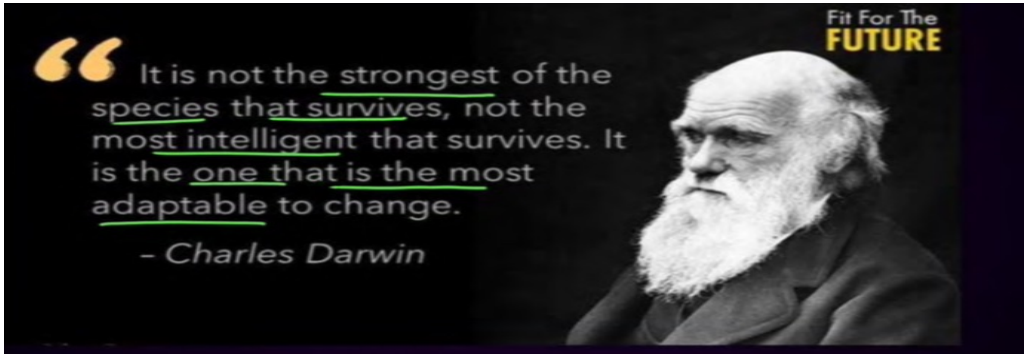
- proposed by a French naturalist Lamarck
- According to this theory:
- 1) all organisms have internal vital force that enable them to develop a new organ or modify the existing organ
- 2) change in environment, force the organism to develop a new organ or modify the existing
- 3) the more we use an organ, the more it develops, the less we use an organ, it disappears (use, and disuse)
- Note: vestigial organ like wisdom tooth, appendix may have a little support to this.
- The character acquired during the lifetime is also passed under the next generation

Lamarck gave example of giraffe to support his theory

- he says- giraffe, developed longer neck over time due to foraging leaves on tall trees when grass was not available
- Note: Weismann theory of germplasm disproved the theory of inheritance of acquired character, he cut the tail of mice for 22 generation, still it appeared
- Two types of cells:
- 1) somatic
- 2) germ cell- any change in germplasm pass to next generation







Darwinism/ theory of natural selection

- Darwin, born in England, went on a voyage on HMS (her Majesty ship) beagle to Galapagos Island located in Pacific Ocean
- At the same time, Alfred Russel Wallace, travelled to Malay Archipelago located in the Indian Ocean
- Both came up with similar conclusion for evolution of species
- Darwin published a book ' origin of species' mentioning theory of natural selection

Darwin was greatly influenced by ' population theory' by Thomas

- population has a tendency to grow exponentially, but population size is stable, more or less, except seasonal fluctuation
- Points of Darwin's theory
- 1) over production or differential reproductive- there is tendency for organisms to overproduce (eg- dogs- 5/6 puppies, and different organisms reproduce at different rate)

- 2) population is stable
- 3) resources are limited- competition and struggle for existence

① Intraspecific — b/w the members of same species

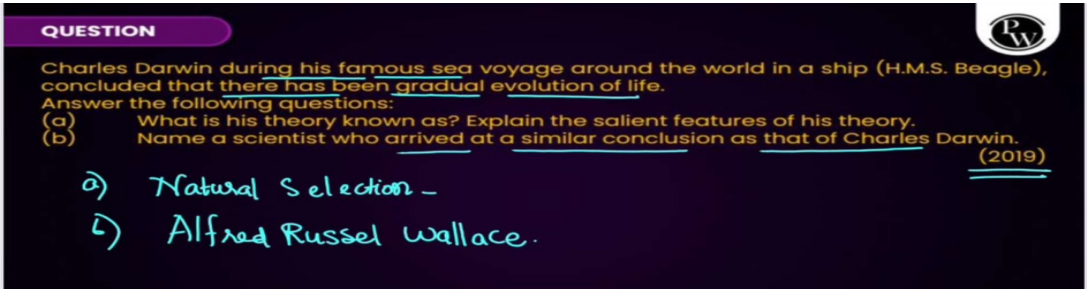
② Interspecific — b/w the members of different species

③ environmental — cold, rain, hot temp, etc

- 4) variations- no two individuals in a species are same, variations exists, useful variations, which, according to Darwin is adaptation, is advantageous and enable the organism to survive
- Variation small, directional and accumulate over time generation after generation. Eg- polar bear in cold climate., Camel in desert
- 5) nature select individuals with advantageous variation- which now reproduces more, leaves more progeny and other eliminate (reproductive fitness)

Survival of fittest, term by Herbert Spencer

- natural selection (tells about mechanism of evolution)
- Branching descent (pattern of evolution)- it shows how all species are related to common ancestry



Mutation theory

- by Hugo devries
- he worked on evening, Primrose (Oenothera lamarckiana)
- Single step, large mutation- saltation

↓

Mutation are responsible for evolution

↓

Large, random and directionless occur by chance

↓

Unpredictable

Modern synthetic theory

- various factors together are responsible for evolution
- 1) natural selection
- 2) mutation
- 3) genetic recombination
- 4) genetic drift
- 5) gene migration

Example examples of natural selection

1) Darwin's finches
2) industrial melanism
3) antibiotic resistant microbe/ bacteria
4) pesticide resistant bacteria

Industrial melanism

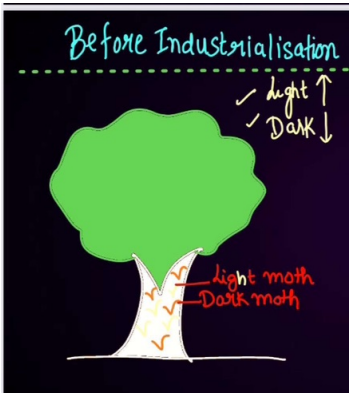
- before industrialisation (1850) in England
- White winged moth (biston betularia) were increasing in numbers, than dark winged moth (biston carbonaria) . No pollution, (no industries.)

↓

Lichens covered bark of tree

↓

Here, white winged moth could easily camouflage in light background due to lichens whereas, dark winged due to its contrasting colour could easily predated by bird



- after industrial (1920)

↓

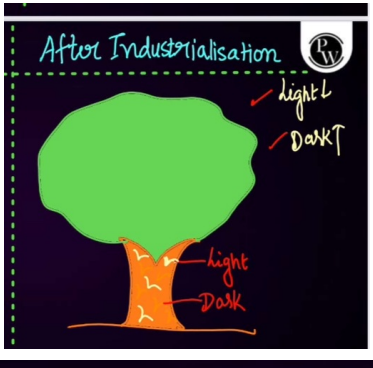
Lichens disappeared

↓

Bark turned darker due to pollution

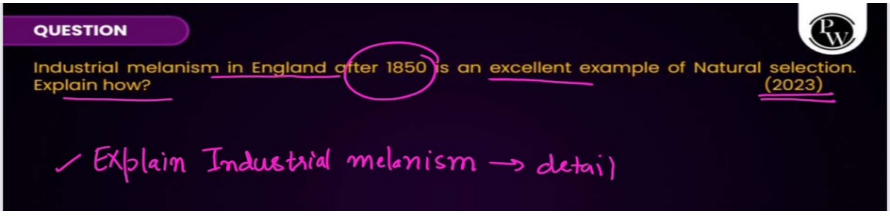
↓

White winged moth easily predicted in dark background, whereas dark winged moth survived



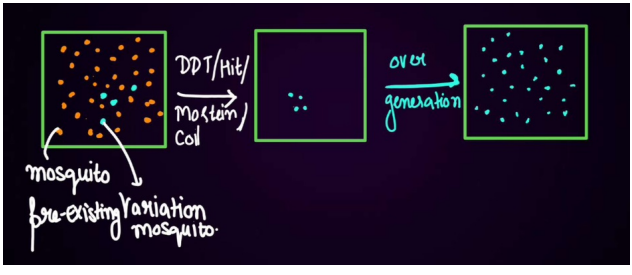




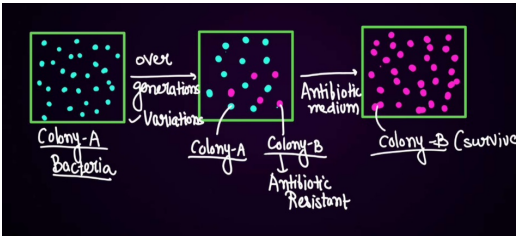




Pesticide resistant mosquito



Microbial experiment

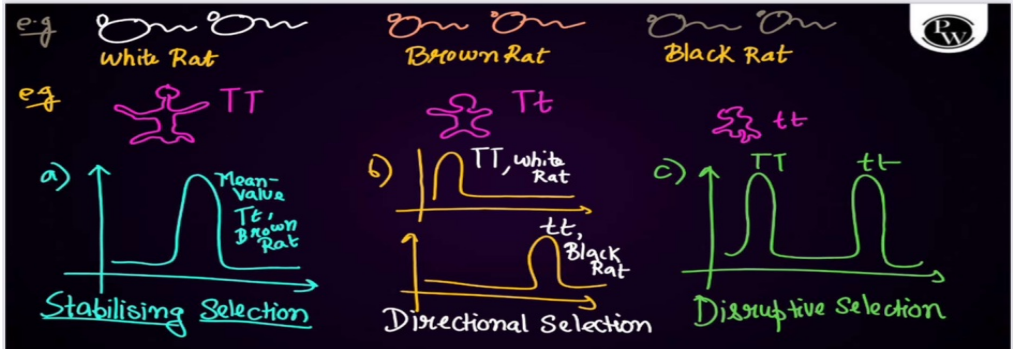


Note:

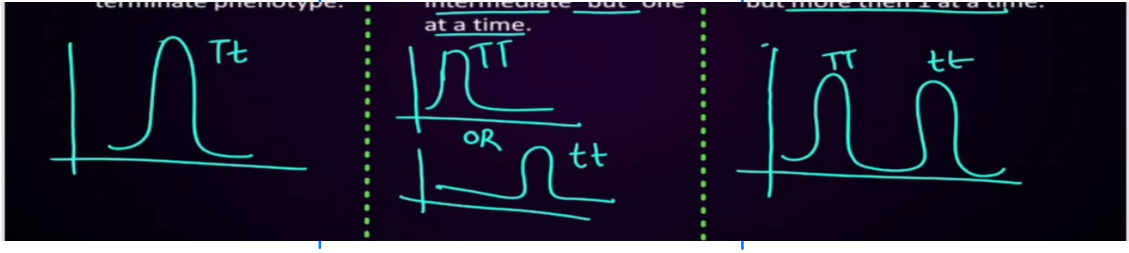
- rate of appearance of new life forms is directly linked to the lifespan/ life cycle of organism
- Microbe divided very fast, hence the results are obtained within few hours (result- new species)
- For the same to occur in a fish etc, it will take million of years as their lifespan is in years

Types of natural selection

1) stabilising selection/ normal selection
2) directional selection/ progressive selection
3) disruptive selection



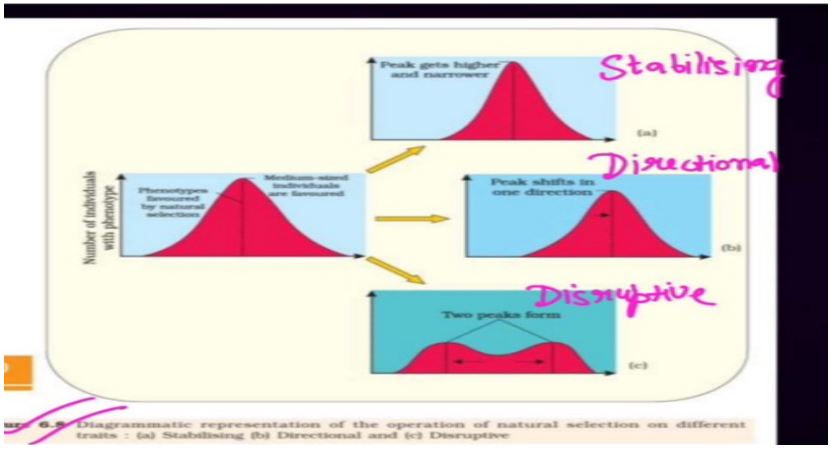
Stabilising	Directional	Disruptive
In this, nature selects the average mean value character or Intermediate phenotype	Here, extreme / peripheral/ values are selected not intermediate but one at a time	Here, extreme/ peripheral/ values are selected not intermediate but more then 1 at a time



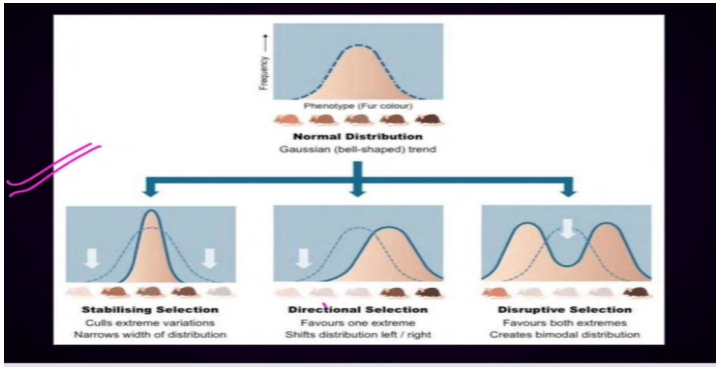
- eg: Shells of snails (Mu[redacted]
- Black → camouflage in Rock
- Grey → cannot camouflage
- White → camouflage in sand

→ Both Selected

eg of Disruptive Selection



Diagrammatic representation of the operation of natural selection on different traits : (a) Stabilising (b) Directional and (c) Disruptive



QUESTION

Explain the three different ways in which the natural selection operates. (2019 C)

1) Stabilising
2) Directional
3) Disruptive

Evidence of evolution

- morphological, and anatomical evidences
- Paleontological evidence
- Embryological evidences
- biochemical evidences
- Artificial selection

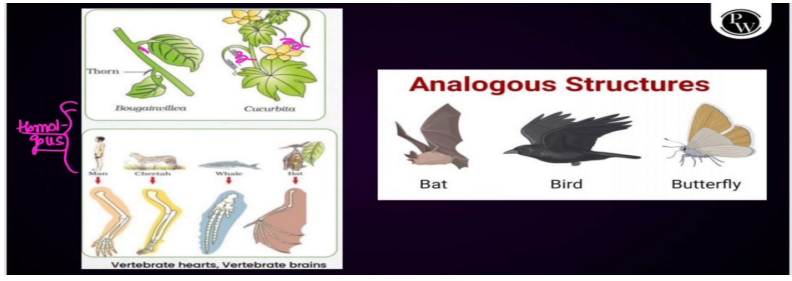
Morphological and anatomical evidences

Homology/ homologous structure

- structures with similar basic plan or common origin, but perform different function
- Homology is a result of divergent evolution
- Even after having common origin due to different geographical areas and different adaptation due to different needs, they Are functionally diverged
- All have similar bones radius, ulna, humerus, carpals, metacarpals, but still perform different function.
- Examples of homo of a structure
- 1) thorns of Bougainvillea and tendrils of Cucurbita (both have modified auxiliary bud).
- 2) Vertebrates heart is 2/3/4 chambered
- 3) Vertebrates brain-forebrain, mid brain, hindbrain

Analogous structure

- structures having different origin or anatomy, but performing similar function
- Analogy is a result of convergent evolution
- Even after having different origin in the structure, evolve functionally same, due to similar environment and similar needs
- Example of analogue structure
- 1) wings of bird and butterfly
- 2) eyes of octopus and eyes of mammals
- 3) sweet potato, and potato
- 4) flippers of penguin and dolphin



QUESTION

(a) Differentiate between analogous and homologous structures.
(b) Select and write analogous structures from the list given below:
(i) Wings of butterfly and birds
(ii) Vertebrate hearts
(iii) Tendrils of Bougainvillea and Cucurbita
(iv) Tubers of sweet potato and potato

(2018)

b) 1, 4 → Analogous.

QUESTION

Select two pairs from the following which exhibit divergent evolution. Give reasons for your answer.
(a) Forelimbs of cheetah and mammals
(b) Flippers of dolphins and penguins
(c) Wings of butterflies and birds
(d) Forelimbs of whales and mammals

Homologous

(AI 2015)

1, 4 → Homologous

QUESTION

Analogous organs are a result of convergent evolution. Justify with the help of suitable example for each.

(2020)

QUESTION

Homologous organs indicate

(2024)

A convergent evolution
B divergent evolution
C adaptive radiation
D natural selection.

QUESTION

Wings of birds and wings of butterflies contribute to locomotion. Explain the type of evolution such organs are a result of. (2020)

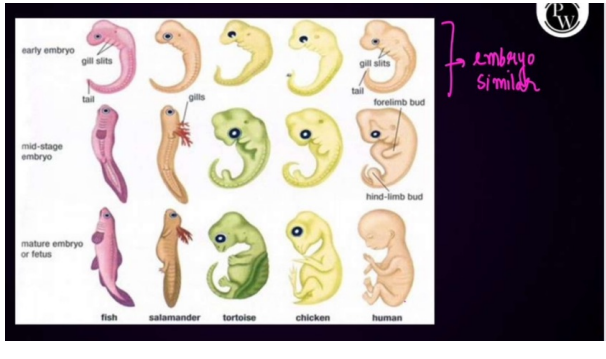
Analogous

Biochemical evidences

- similarities in protein and genes performing a given function among diverse organisms, give clues to common ancestry
- Example- sequence similarities in DNA between chimpanzee and human
- Similar enzyme trypsin found in vertebrate and invertebrate

Embryological evidences

- Ernst Haeckel gave - biogenetic law/ recapitulation theory
- According to this ' ontogeny recapitulates phylogeny'
- Certain features during embryonic stage, common to all vertebrates that are absent in adult
- For example, the embryos of all vertebrates, including human develop a row of vestigial Gill slits just behind the head, but it is a functional organ only in fish and not found in any other adult vertebrates.
- Karl von Baer disproved recapitulation theory, and said embryos, never passed through adult stage of other organism (ancestor)



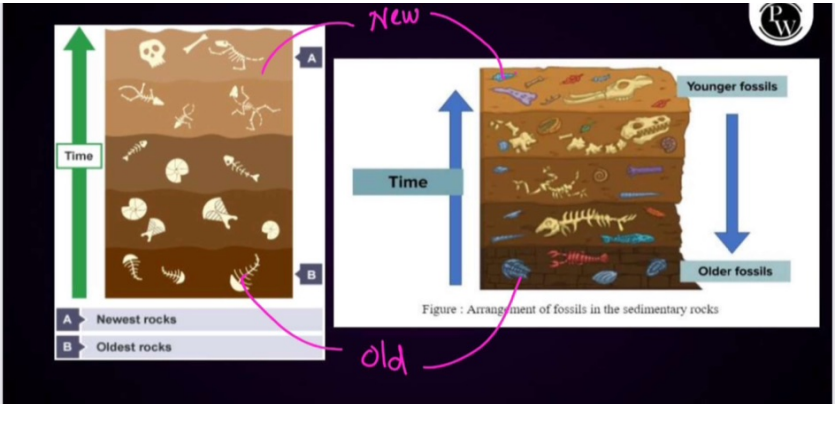
QUESTION

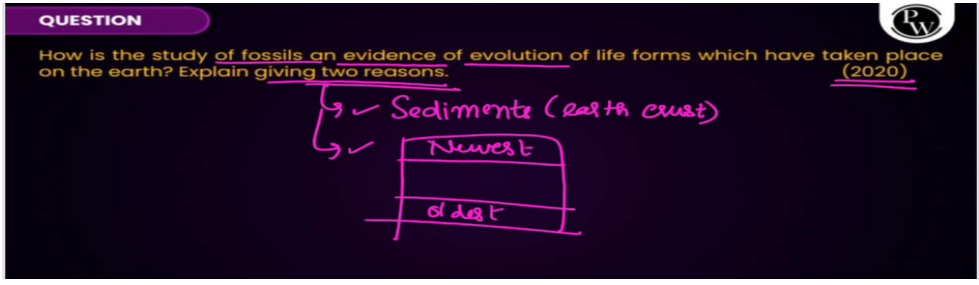
The scientist who proposed that "embryos of all vertebrates including humans develop a row of vestigial gill slits behind the head but it is a functional organ only in fish and replaced by lungs in all other vertebrates" is (2024)

A. Darwin
B. Lamarck
C. Ernst Haeckel
D. Mendel.

Paleontological evidences

- fossils- are the complete organism or its part or footprint or excreta preserved over million years (extinct)
- Paleontoology- study of fossil
- Rocks and sediments - The cross-section of earth's crust indicates the arrangement of sediments one over the other during the long history of earth
- Different age rocks, sediments contain fossils of different life forms who died during the formation of the particular sediment
- They represent extinct organisms (eg- dinosaur)
- a study of fossils in different sedimentary layers indicates the geological period in which they existed
- Age of the fossil can be determined by:
- 1) relative dating- comparing the sediments
- 2) radioactive dating- carbon dating, etc

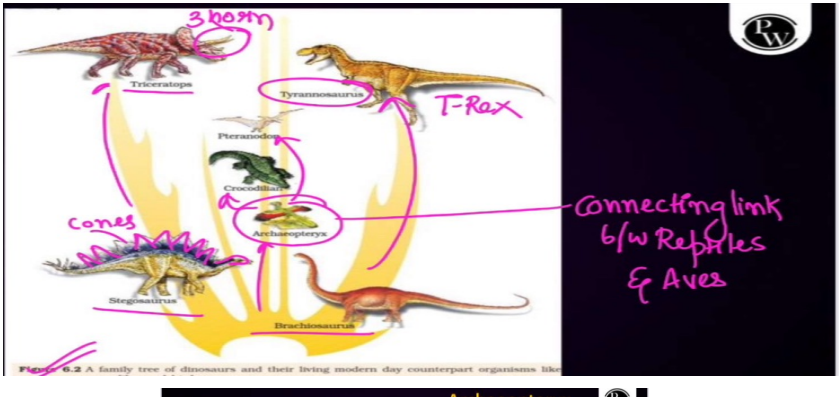


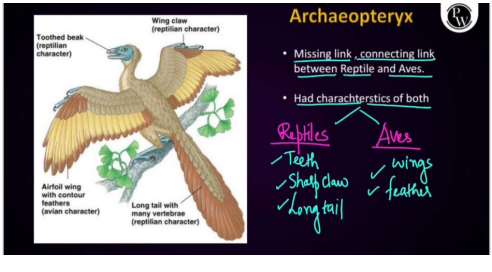


Artificial selection

- as an evidence of evolution
- a process where humans choose organisms with desirable traits and breed them to create new organisms with those traits
- Eg- in agriculture, horticulture, etc

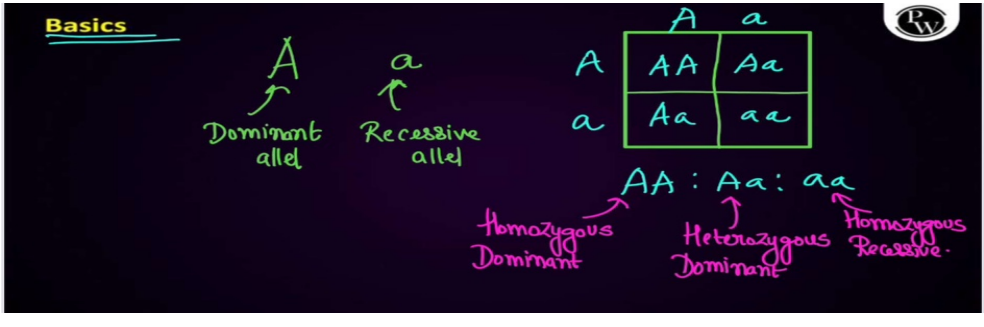


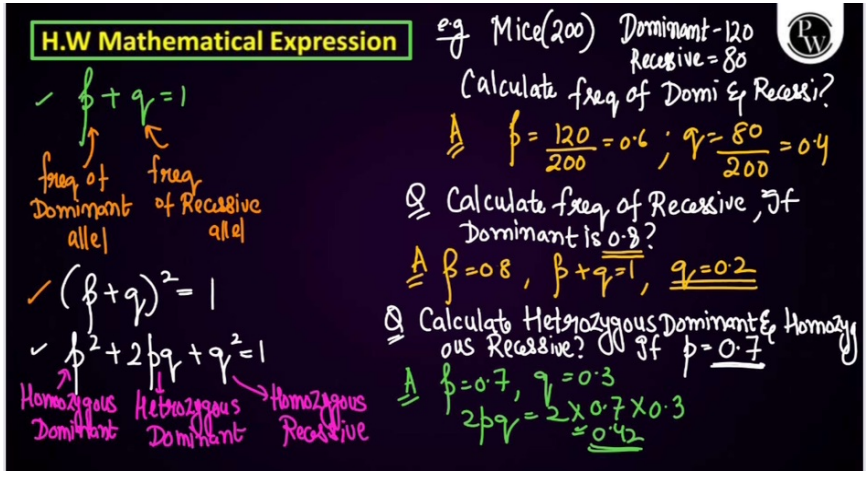




Hardy Weinberg principle

- allele- alternate form of gene
- gene pool - sum total of gene in a population
- HW principle - in a large randomly mating population , allelic frequency remain constant generation after generation if the population is free from evolutionary forces
- Factors affecting:
- 1) gene migration
- 2) genetic recombination
- 3) gene mutation
- 4) genetic drift
- 5) natural selection





QUESTION

A population is in genetic equilibrium/Hardy- Weinberg equilibrium for a gene with 2 alleles (dominant allele is 'A' and recessive allele 'a'). If the frequency of allele 'A' is 0.6, then the frequency of genotype 'Aa' is (2024)

A 0.21

B 0.48

C 0.42

D 0.32.

Heterozygous Dominant

A = 0.6

Aa = ?

A = 0.6

a = 0.4

$2pq = 2Aa = 2 \times 0.6 \times 0.4 = 0.48$

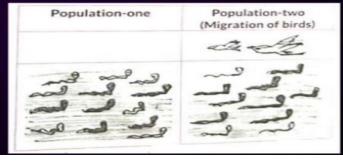
QUESTION

A population of snakes lived in a desert with brown sand. Study the drawings given below showing the change in the population from 'one' to 'two' over time and answer the question that follows. Brown snakes and grey snakes are represented by alleles A/a (Dominant/recessive).

(a) If the frequency of the recessive trait is 9% in population-one, work out the frequency of homozygous dominant and heterozygous dominant snakes.

(b) Name the mechanism of evolution that must have operated so that population-two evolved from population-one. (2024)

a = 9%



a) $a = \sqrt{0.09} = 0.3$

$A = 0.7$

$2pq = 2 \times 0.7 \times 0.3 = 0.42 = 42\%$

QUESTION

$p^2 + 2pq + q^2 = 1$. Explain the algebraic equation on the basis of Hardy Weinberg's principle. (Delhi 2017)

$p^2 + 2pq + q^2 = 1$

- $p = ?$
- $q = ?$
- $2pq = ?$

QUESTION

Hardy-Weinberg principle is stated in the following algebraic equation:

$$p^2 + 2pq + q^2 = 1$$

(a) State what do 'p' and 'q' denote in the equation.

(b) State Hardy-Weinberg principle as indicated in the equation.

(c) What would you interpret if the value of 1 in the equation gets deviated? (2020)

$p + q = 1$

↑ Stable

Factors affecting HW equilibrium

1) Gene migration- if multiple migration occurs in a population

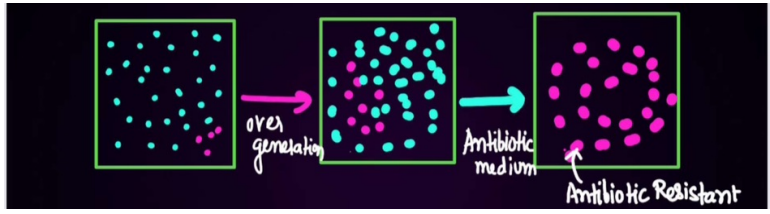


2) genetic recombination- during crossing over, in gamete formation



3) mutation- lederberg replica plate experiment

- pre-existing advantageous mutation when selected will form a result in New phenotype
- Over generation- new species formed



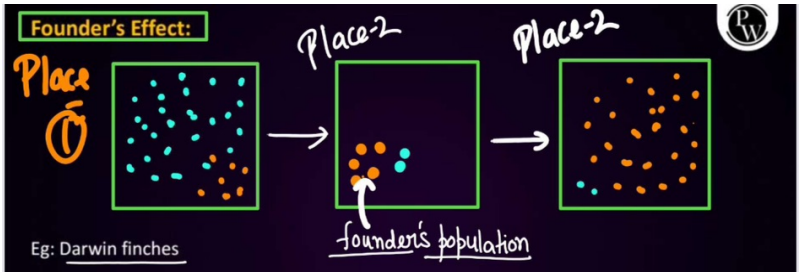
4) natural selection

5) genetic drift- in a small population, random change of allele maybe due natural calamity, anthropogenic action(human action), geographical isolation(separate area)

- two types of genetic drift. → Founder's effect.
  → Bottleneck effect

Founder's effect

- eg- Darwin's finches
- There is an original population of seed, eating bird living in mainland(South America) , some of these moved to Galapagos Island, these are called founders population
- as per their feeding habits, their beak modified with time, some became insect, some cactus eating, nectar eating, fruit eating, etc



Bottleneck effect

- initial population was filling the complete, but due to hunting, only a few left , size of the neck



QUESTION

(a) How does the Hardy-Weinberg equation explain genetic equilibrium?
(b) Describe how this equilibrium is disturbed that may lead to founder effect. (AI 2019)

a) $p+q=1$
b) factors affection

QUESTION

Assertion (A): Gene flow increases genetic variations.
Reason (R): The random introduction of new alleles into a recipient population and their removal from donor population affects allele frequency. [Gene migration] (2024)

A. Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

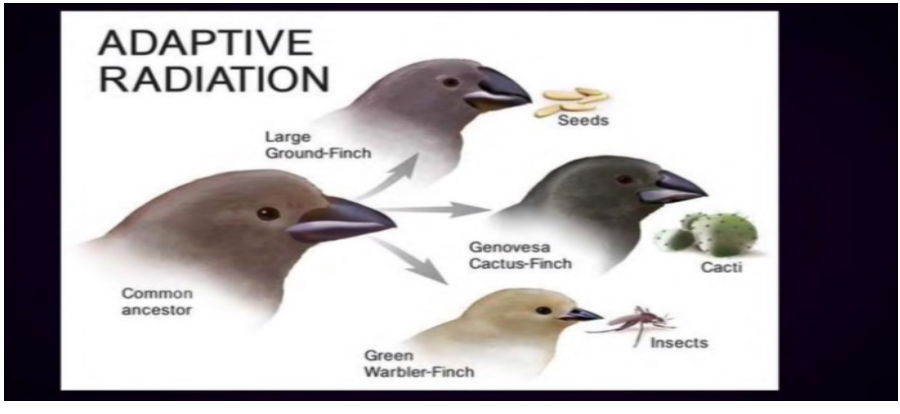
B. Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

C. Assertion (A) is true, but Reason (R) is false.

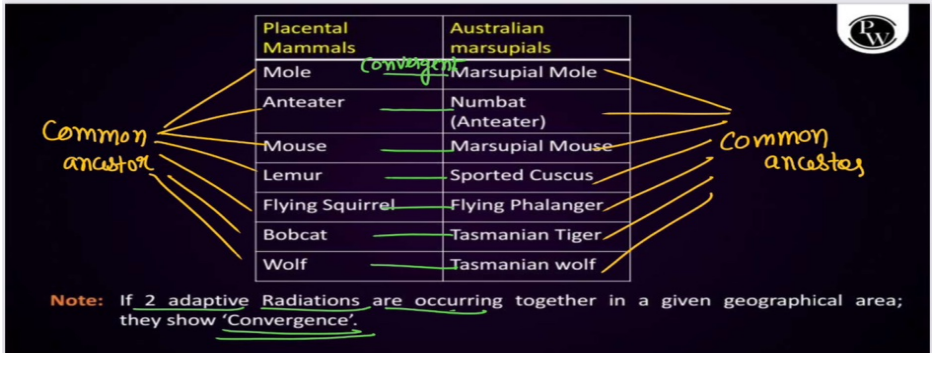
D. Assertion (A) is false, but Reason (R true.

Adaptive radiation

- the evolution of different species, starting from a common geographical point radiating along various direction
- Eg- Darwin's finches. Their beak modified, depending upon the food, they explored
- Adaptive radiation indicates divergent evolution
- Other example-placental mammals, Australian marsupial







QUESTION

Identify the option that gives the correct type of evolution exhibited by the two animals shown, living in the same habitat in Australia. (2023)

A Convergent evolution

B Disruptive selection

C Divergent evolution

D Homologous ancestry



QUESTION

(a) Identify any two marsupials from the list given below:
(i) Lemur phalanger,
(ii) Spotted cuscus,
(iii) ~~Flying~~
(iv) Bobcat,
(vi) Mole,
(v) Tasmanian wolf

(b) "Australian marsupials exhibit adaptive radiation. Justify the statement. (2020)

↳ Table

QUESTION

(a) How did Darwin explain adaptive radiation?
(b) Give another example exhibiting adaptive radiation. (3/5, Delhi 2016)

↳ e.g Darwin finches, PM, AM.

Geological time scale

Geological time scale

✓ Eon ✗
↓
Era
↓
Period → NCERT
↓
Epoch ✗

Newest ↑ oldest

ERA	Period
• Coenozoic	• Tertiary • Quaternary
• Mesozoic	• Cretaceous • Jurassic • Triassic
• Paleozoic	• Permian • Carboniferous • Devonian • Silurian • Ordovician • Cambrian
• Precambrian/ Proteozoic	

- universe- 20 bya
- Earth- 4.5 bya
- Invertebrates- 500 mya
- Jawless fish- 350 mya
- seaweed, and early land plants- 320 mya

- we are told that the first organism that invaded land were plants
- Fish with stout and strong fins
- Could move our land and go back to water- 350 mya

- in 1938, a fish caught off in South Africa happened to be a coelacanth which was thought to be extinct. These animals called lobe fins evolved into the first amphibians that lived on both land and water

- the amphibians evolved into reptile
- in the next 200 million years or so, reptiles of different shapes and sizes dominated on earth
- Some of these land reptiles went back into water to evolve into fishes like reptiles- 200 mya (Ichthyosaurs)
- The land reptile were, of course the dinosaurs. The biggest of them I.e, Tyrannosaurus Rex was about 20 feet in height and had a huge fearsome dagger like teeth
- Dinosaur suddenly disappeared from the Earth- 65mya
- The first mammals were like shrews. Their fossils are small sized. when reptiles Declined, Mammals took over this earth.
- note: initially, there was a common land (Pangea)

↓

Continental drift

↓

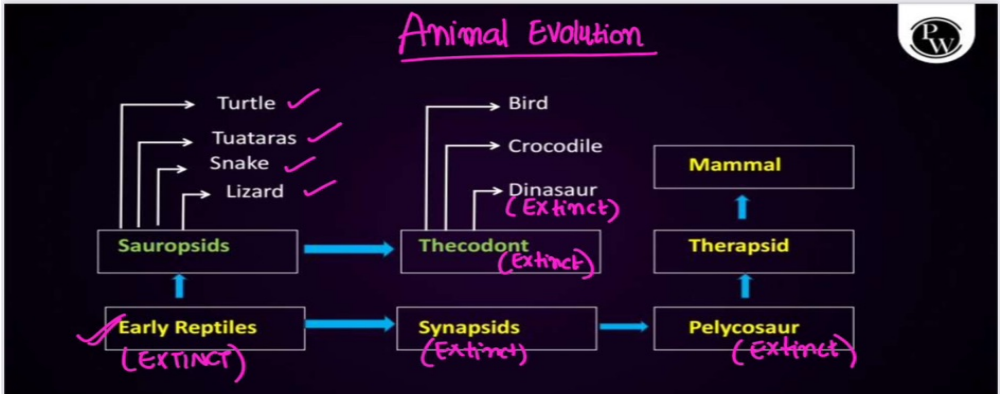
Land separated

↓

Due to this, South America joined North America. Earlier there were mammals like hippopotamus, horse, rabbit, etc in South America, but later when it joined North America, North America fauna dominated

- in Australia, pouched mammal survived due to no other mammal dominated





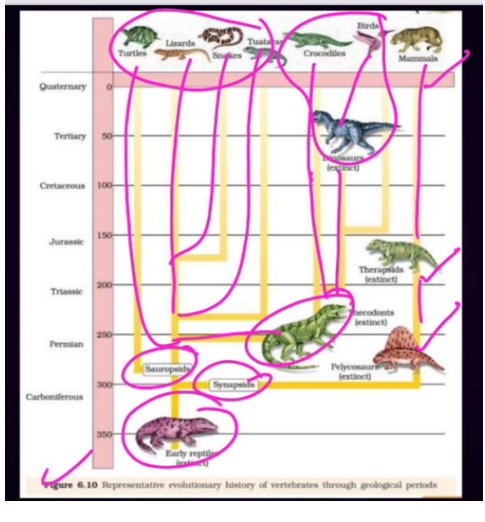
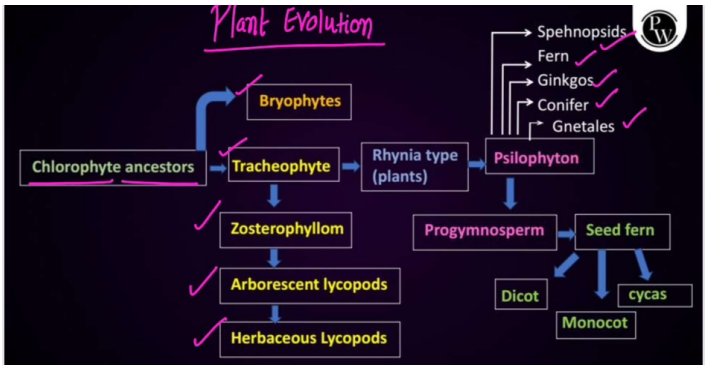


Figure 6.10 Representative evolutionary history of vertebrates through geological periods



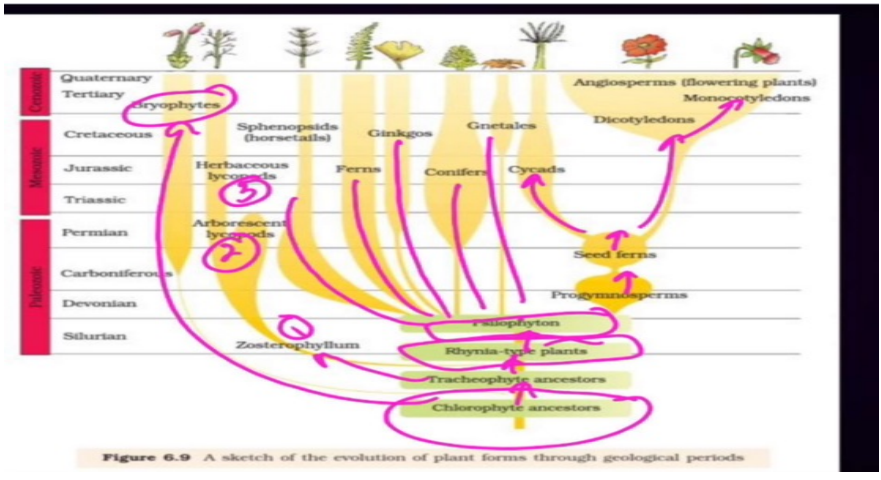
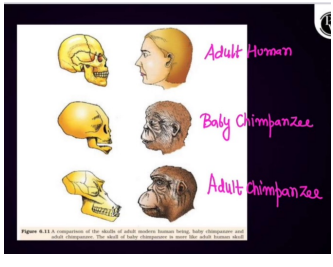


Figure 6.9 A sketch of the evolution of plant forms through geological periods

Human evolution

- mammals. → Primate. → Monkey
  → Apes- chimpanzees, gorillas
  → Human

note- when we look at this skull of human and baby and adult chimpanzee, baby chimpanzee looks similar to adult human skull, than adult chimpanzee skull, showing common ancestry



1) Dryopithecus
2) Ramapithecus
3) Australopithecus
4) Homo Habilus
5) Homo Erectus
6) Homo Neanderthal
7) Cromagnon
8) Homo Sapiens

TRICK

Dryopithecus ----------.	Doctor
Ramapithecus----------.	Ram in
Australopithecus--------.	Australia
Homo Habilis----------.	Have researched
Homo Erectus----------.	Evolution on
Homo Neanderthal----------.	New
Homo Sapiens----------.	Homo Sapiens

1. Dryopithecus
- 15 mya, direct ancestor of ape
- hairy body, more ape like

2. Ramapithecus
- 15 mya
- more human like
- Hairy, fossils obtained from Shivalik hill India

3. Australopithecus
- 500 cc (Cubic Centimeter) cranial capacity
- First time Bipedal movement
- 4 feet height, lived in east African grassland.
- Fossils obtained from Ethopia & Tanzania
- Made stone weapon to eat Fruit.

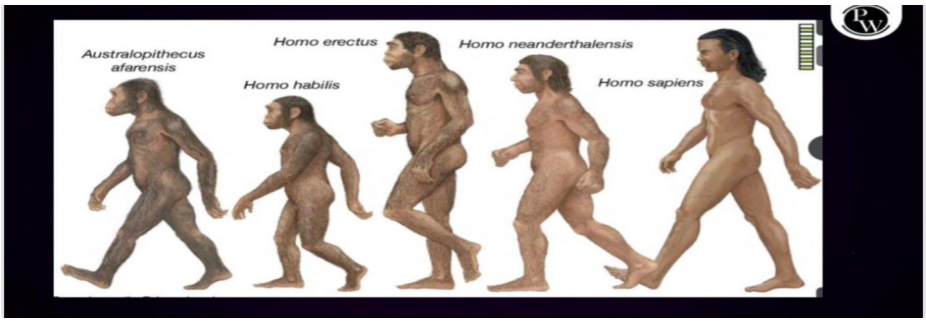
4. Homo habilis
- Hominid like human
- Able - Man
- Perfect tool maker
- Probably did not eat meat
- 650-800 cc cranial capacity

5.Homo errectus
- 900 cc cranial capacity
- First time erect posture
- Used fire, Probably ate meat
- Fossils obtained, 1891, from JAVA

6. Neanderthal
- Probably lived in central & east asia – 4000-1 lakh yr ago
- 1400 cc
- first time covered their body & buried dead one & put flower on them

7.Cromagnon
- Not in NCERT
- 1650 cc
- Direct ancestory of today's modern
- Made cave painting
- 18000 yr ago, a cave painting was obtained from Bhimbetka, raison district of M.P

8.Homo sapien sapien
- 75000-10,000 yr ago
- ICE-AGE is when they arose
- Brought agriculture revolution
- 10,000 year ago
- Human settlement, industrial revolution
- 1350 cc



QUESTION

(a) Rearrange the following in the correct order of their appearance on earth between two million years and 40,000 years back.
Neanderthals, Australopithecus, Homo erectus and Homo habilis.

(b) Which one of the above
(i) had the largest brain size
(ii) ate fruits?

(2020)

Aust — Neanderthal

a) Australo → Homo habilus

QUESTION

At which stage during evolution did human use hides to protect their bodies and buried their dead? (2023)

A. Homo habilis
B. Neanderthal man
C. Java man
D. Homo erectus

QUESTION

Write two differences between Homo erectus and Homo habilis. (2/3, Delhi 2019)

Homo Habilus

Name the first human like hominid. Mention his food habit and brain capacity. (AI 2015 C)

Write the probable differences in eating habits of Homo habilis and Homo erectus. (AI 2016)

Write the names of the following:
(a) A 15 mya primate that was ape-like. → Dryopithecus
(b) A 2 mya primate that lived in East African grasslands. (Australopithecus) (2018)